

CO1-AT2 – Agile Sprint Planning & User Story Workshop

Problem Statement: Smart Smart Grid Cybersecurity Management Platform

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Methodology: The Agile Scrum Approach

This sprint planning document applies the Agile Scrum methodology to the Smart Grid Cybersecurity Management Platform. Scrum is selected because cybersecurity systems require continuous monitoring, frequent software updates, and rapid responses to emerging cyber threats. Unlike traditional software projects, cybersecurity applications must adapt quickly to evolving attack techniques, making Agile Scrum an ideal development methodology. The project is divided into short development cycles called sprints, where each sprint focuses on implementing, testing, and improving one or more cybersecurity features.

The Scrum framework enables continuous collaboration among developers, cybersecurity analysts, network administrators, utility companies, and other stakeholders. Product requirements are prioritized in the Product Backlog, selected tasks are moved into the Sprint Backlog, and development is carried out during each sprint. Daily Scrum meetings help monitor progress, while Sprint Reviews and Retrospectives ensure continuous improvement of both the software and development process.

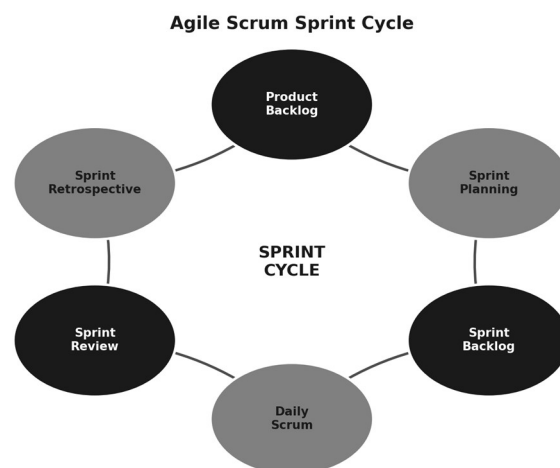


Figure 1: Agile Scrum sprint cycle followed for this workshop

1. Project Vision and Stakeholders

1.1 Project Vision

The vision of the Smart Grid Cybersecurity Management Platform is to build an intelligent security solution that continuously protects smart grid infrastructure from cyber threats through Artificial Intelligence, Machine Learning, and real-time monitoring technologies. The platform aims to detect malicious activities, predict future cyberattacks, monitor communication networks, generate compliance reports, and provide instant alerts to security teams. By

integrating advanced cybersecurity technologies into a centralized platform, the system improves the reliability, resilience, and overall security of modern smart power grids while ensuring uninterrupted electricity distribution.

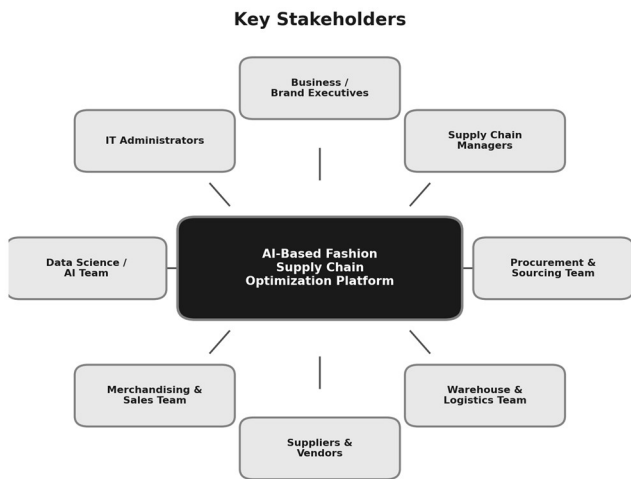
1.2 Project Objectives

- Forecast product demand using historical sales, seasonality, and trend data.
- Optimize inventory allocation across warehouses and retail stores.
- Coordinate suppliers for timely, reliable replenishment.
- Track shipments in real time across the logistics network.
- Monitor sales trends to detect fast- and slow-moving products.
- Generate consolidated supply chain performance reports.

1.3 Key Stakeholders

The platform serves a cross-functional group of stakeholders, each with distinct goals and pain points that the product backlog must address:

Stakeholder	Role in the Project	Primary Interest
Business / Brand Executives	Set strategic direction; review supply-chain KPIs	Profitability and brand growth
Supply Chain Managers	Own inventory allocation and overall coordination	Balanced stock levels, fewer disruptions
Procurement & Sourcing Team	Manage supplier relationships and purchase orders	Reliable, cost-effective sourcing
Warehouse & Logistics Team	Receive, store, and dispatch inventory; track shipments	Accurate, real-time shipment visibility
Suppliers / Vendors	Fulfil purchase orders and share production status	Clear demand signals and lead times
Merchandising & Sales Team	Interpret sales trends; plan assortments and promotions	Early trend detection
Data Science / AI Team	Build and maintain forecasting and optimization models	Access to clean, timely data
IT Administrators	Maintain platform infrastructure, security, and integrations	System reliability and uptime



2. Epics and User Stories

Monitor smart grid security continuously across all connected devices and communication networks.

- Detect cyber threats such as malware, ransomware, phishing attacks, unauthorized access, and denial-of-service attacks using AI-based detection techniques.
- Predict future cyberattack patterns using Machine Learning models trained on historical cybersecurity data.
- Track network activities in real time to identify abnormal communication behavior and potential security vulnerabilities.
- Generate automated cybersecurity compliance reports based on government regulations and industry standards.
- Notify cybersecurity teams immediately through alerts, emails, SMS, and dashboard notifications whenever suspicious activities are detected

Epic 1 – Demand Forecasting

ID	User Story
US-1.1	As a merchandising manager, I want the system to forecast product demand from historical sales and seasonal trend data, so that I can plan purchase orders accurately.
US-1.2	As a supply chain manager, I want demand forecasts broken down by region and season, so that I can allocate inventory accordingly.

Epic 2 – Inventory Allocation Optimization

ID	User Story
US-2.1	As a warehouse manager, I want the system to recommend optimal stock allocation across stores, so that inventory matches regional demand.
US-2.2	As a supply chain manager, I want alerts for understocked or overstocked items, so that I can take corrective action quickly.

Epic 3 – Supplier Coordination

ID	User Story
US-3.1	As a procurement manager, I want to view supplier performance metrics, so that I can select reliable partners.
US-3.2	As a procurement officer, I want to send automated purchase orders to suppliers, so that replenishment cycles are faster.

Epic 4 – Shipment Tracking

ID	User Story
US-4.1	As a logistics coordinator, I want to track shipments in real time, so that I can anticipate delays.
US-4.2	As a warehouse manager, I want automated notifications when shipments arrive, so that I can plan receiving staff.

Epic 5 – Sales Trend Monitoring

ID	User Story
US-5.1	As a merchandising analyst, I want to monitor real-time sales trends by category, so that I can identify fast-moving products.
US-5.2	As a marketing manager, I want alerts on emerging fashion trends, so that I can plan promotional campaigns.

Epic 6 – Supply Chain Reporting

ID	User Story
US-6.1	As a supply chain executive, I want automated supply chain performance reports, so that I can review KPIs regularly.
US-6.2	As a business owner, I want a summary dashboard, so that I can make quick strategic decisions.

3. Acceptance Criteria

Acceptance criteria define the measurable conditions under which each user story is considered complete. The Given–When–Then format is used below for the highest-priority user story from each Epic.

Story	Given	When	Then
US-1.1	the system has at least 12 months of historical sales data	a merchandising manager requests a demand forecast for a product	the system displays a forecast with a confidence interval for the next 4–12 weeks
US-2.1	current stock levels and regional sales data are available	the allocation engine runs its recommendation cycle	it proposes a store-wise allocation plan that reduces projected stock-outs by a target threshold
US-3.2	an approved purchase order exists in the system	the procurement officer confirms the order	the order is transmitted to the supplier automatically and a confirmation is logged
US-4.1	a shipment has been dispatched with a tracking reference	its location status changes	the dashboard updates the shipment's status within 15 minutes
US-5.1	point-of-sale data is streaming into the platform	sales of a category exceed the defined trend threshold	the system flags the category as “trending” and notifies the merchandising team
US-6.1	a reporting period has ended	the scheduled report job runs	a supply chain performance report (KPIs, stock levels, supplier delays) is generated and emailed to executives

4. Product Backlog Prioritization

The backlog is prioritized using the MoSCoW technique (Must have, Should have, Could have, Won't have this sprint), weighed against business value and dependency on other stories. Forecasting and allocation stories are prioritized first because every downstream capability — supplier orders, shipments, and reporting — depends on accurate demand and stock data.

ID	User Story (Summary)	Priority	Business Value	Dependency
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US-1.1	Forecast demand from historical/seasonal data	Must	High	None (foundational)
US-2.1	Recommend optimal stock allocation	Must	High	Depends on US-1.1
US-2.2	Alert on under/overstocked items	Must	High	Depends on US-2.1
US-3.2	Send automated purchase orders	Should	High	Depends on US-2.2
US-4.1	Real-time shipment tracking	Should	Medium	Depends on US-3.2
US-1.2	Forecasts by region and season	Should	Medium	Depends on US-1.1
US-3.1	Supplier performance metrics	Could	Medium	Independent
US-5.1	Monitor sales trends by category	Could	Medium	Independent
US-4.2	Shipment-arrival notifications	Could	Low	Depends on US-4.1
US-6.1	Automated performance reports	Won't (Sprint 1)	High	Depends on most modules

5. Sprint Backlog Planning — Sprint 1

Sprint 1 focuses on building the core cybersecurity capabilities required for protecting the smart grid infrastructure. The selected user stories establish the foundation for monitoring, detecting cyber threats, and generating security alerts. These modules will later support advanced AI prediction, compliance reporting, and security analytics. Each user story is divided into manageable development tasks assigned to the appropriate development teams.

6. Story Point Estimation

Story points are estimated using the **Planning Poker** technique with the **Fibonacci sequence (1, 2, 3, 5, 8, 13)**. The estimates represent the relative effort, complexity, technical risk, and uncertainty involved in implementing each user story rather than the actual development time. Higher story points indicate greater complexity and additional development effort. The estimation process helps the Scrum team balance the sprint workload and improve project planning.

Figure 4: Fibonacci scale used for estimation

ID	User Story (Summary)	Points	Justification
ID	User Story (Summary)	Story Points	Justification
US-1.1	Monitor Grid Security	8	Requires continuous monitoring, dashboard development, and integration with smart grid devices.
US-1.2	Real-time Security Dashboard	5	Requires dashboard visualization and live data updates.
US-2.1	Detect Cyber Threats	8	AI-based threat detection requires machine learning models and large security datasets.
US-2.2	AI Threat Classification	5	Extends the detection module with threat classification algorithms.
US-3.1	Predict Attack Patterns	8	Involves predictive analytics and historical cyberattack analysis.
US-3.2	Predictive Security Reports	5	Builds reports using prediction results.
US-4.1	Track Network Activities	5	Requires continuous monitoring of communication traffic.

US-4.2	Maintain Security Logs	3	Log management is rule-based and moderately complex.
US-5.1	Compliance Reports	5	Generates automated reports using collected security logs.
US-5.2	Compliance Dashboard	3	Displays summarized compliance information.
US-6.1	Instant Security Alerts	3	Event-driven notification service with email and SMS integration.
US-6.2	Multi-channel Notifications	2	Extends the alert system with additional communication methods.

Sprint 1 committed capacity: US-1.1 (8) + US-2.1 (8) + US-2.2 (3) = 19 story points, judged achievable within a two-week sprint for the assigned team size.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

By the end of Sprint 1, deliver a working demand-forecasting and inventory-allocation module that produces reliable stock recommendations and flags under/overstocked items — establishing the data foundation that all later Epics (supplier coordination, shipment tracking, and reporting) will build upon.

7.2 Sprint 1 Deliverables

- A functioning demand-forecasting model exposed through an API.
- A store-wise inventory allocation recommendation engine.
- An alerting mechanism for under/overstocked items.
- A basic UI for viewing forecasts and allocation recommendations.

7.3 Expected Outcomes (Project-Level)

- Reduced stock shortages.
- Lower excess inventory.
- Improved supplier coordination.
- Faster replenishment cycles.
- Higher profitability.

7.4 Sprint 1 Board (Jira-style Snapshot)

Sprint execution is tracked on a Jira-style Scrum board with To Do, In Progress, and Done columns. Each card shows the user-story/task ID, a short description, the assigned story points, and the responsible role — giving the team a real-time, at-a-glance view of sprint progress.

Figure 5: Illustrative Sprint 1 board tracking the committed user stories and tasks

Code of Conduct

I, N.Madhusudhan – 192425303(Name / Reg No), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N.Madhusudhan

Date: 23/07/26